Notes for Tight-Binding Calculation

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I. Matlab

A. 1-D

1. SSH model

The 1st quantized Hamiltonian is

$$H = (A + B\cos k)\sigma_x + B\sin k\sigma_y, \tag{1}$$

and k-space $E(k)$ relation could be calculated. There would be critical point at $A = B$ where the gap closes and re-opens. And the non-trivial criterion is $|A| < |B|$ for mass inversion.

Using

$$\cos k = \frac{e^{ik} + e^{-ik}}{2},\ \sin k = \frac{e^{ik} - e^{-ik}}{2i},$$

the 2nd quantized Hamiltonian is

$$\hat{H} = \sum_k \left(a_k^\dagger, b_k^\dagger\right) H \begin{pmatrix} a_k \\ b_k \end{pmatrix} = \sum_k a_k^\dagger (A + Be^{-ik}) b_k + \text{h.c.}. \tag{2}$$

By Fourier transformation,

$$a_k/b_k = \frac{1}{\sqrt{N}} \sum_i e^{-ik\cdot R_i} c_{A/B,i},$$

Hamiltonian in real space is

$$\hat{H} = \sum_i A c_{A,i}^\dagger c_{B,i} + B c_{A,i+1}^\dagger c_{B,i} + \text{h.c.}. \tag{3}$$

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Considering the open boundary condition,

$$\hat{H} = \left(\cdots c^{\dagger}_{A,i+1}, c^{\dagger}_{B,i+1}, c^{\dagger}_{A,i}, c^{\dagger}_{B,i} \cdots\right) \begin{pmatrix} \ddots & & & & \\ & 0 & A & 0 & 0 \\ & A & 0 & B & 0 \\ & 0 & B & 0 & A \\ & 0 & 0 & A & 0 \\ & & & & \ddots \end{pmatrix} \begin{pmatrix} \vdots \\ c_{A,i+1} \\ c_{B,i+1} \\ c_{A,i} \\ c_{B,i} \\ \vdots \end{pmatrix}. \tag{4}$$

When $0 < A < B$ is satisfied, diagonalize the Hamiltonian would give wavefunction with obvious peak at boundary and decay fast when moving inside. And for this model, the winding is given by

$$\nu_1 = \int \omega_1(k)\mathrm{d}k,\ \omega_1(k) = \frac{1}{2\pi}\,\mathrm{Tr}(q^{-1}\mathrm{d}q) = \frac{1}{2\pi}\left[\ln\det(q(\pi)) - \ln\det(q(-\pi))\right],\ q(k) = A + B(\cos k - i\sin k), \tag{5}$$

where $q(k)$ is half of Hamiltonian in Chiral operator's representation. Possible problem here might be that ω would have multivalue at certain k points, especially high symmetry points. And one need to be careful when dealing with such boundary terms.

B. 2-D

1. *Square Lattice*

$$H = \sin k_x \sigma_x + \sin k_y \sigma_y + \left[m - (2 - \cos k_x - \cos k_y)\right]\sigma_z. \tag{6}$$

Using which k-space $E(k)$ could be shown. The non-trivial criterion reads $0 < m < 4$ with $m = 2$ also a phase transition point.

Set the system to $x - y$ plane, to see edge states, one needs to set one direction with open boundary condition in real space while keep the other with periodic condition in k-space. Say that one chooses to open y direction, then one needs to define some *half-Fourier-transformation* to make the ladder operators in the mixing representation $\{k_x, y\}$,

$$a_k/b_k = \sum_i e^{-ik_y \cdot R_{i_y}} a_{k_x,i_y}/b_{k_x,i_y}. \tag{7}$$

Within such representation, one could rewrite the Hamiltonian as

$$\begin{aligned}\hat{H} = &\sum_{k_x,i_y}\left(a^\dagger,\ b^\dagger\right)_{k_x,i_y}\begin{pmatrix} m-(2-\cos(k_x)) & \sin(k_x) \\ \sin(k_x) & -m+(2-\cos(k_x))\end{pmatrix}\begin{pmatrix} a \\ b\end{pmatrix}_{k_x,i_y} \\ &+\sum_{k_x,i_y}\left(a^\dagger,\ b^\dagger\right)_{k_x,i_y}\begin{pmatrix} 1/2 & -1/2 \\ 1/2 & -1/2\end{pmatrix}\begin{pmatrix} a \\ b\end{pmatrix}_{k_x,(i+1)_y} + \text{h.c.}\end{aligned} \tag{8}$$

Which could be written as extended matrix

$$\begin{pmatrix} \ddots & & & & \\ & i+1 & i & i-1 & \\ & [m-(2-\cos(k_x))]\sigma_z+\sin(k_x)\sigma_x & \frac{1}{2}(\sigma_z-i\sigma_y) & 0 & \\ & \frac{1}{2}(\sigma_z-i\sigma_y) & [m-(2-\cos(k_x))]\sigma_z+\sin(k_x)\sigma_x & \frac{1}{2}(\sigma_z-i\sigma_y) & \\ & 0 & \frac{1}{2}(\sigma_z-i\sigma_y) & [m-(2-\cos(k_x))]\sigma_z+\sin(k_x)\sigma_x & \\ & & & & \ddots\end{pmatrix} \tag{9}$$

and could thus be diagonalized numerically. Here $\{i\}_{i=1}^{n}$ denotes number of sites along y direction. And edge states would show themselves in topological area.

To see real space edge wavefunctions surrounding the system, full Fourier transformation is needed,

$$\begin{aligned}\hat{H} = &\sum_{i_x,j_y}(a^\dagger,b^\dagger)_{i_x,j_y}(m-2)\sigma_z\begin{pmatrix} a \\ b\end{pmatrix}_{i_x,j_y} \\ &+\sum_{i_x,j_y}\left(a^\dagger,\ b^\dagger\right)_{i_x,j_y}\frac{1}{2}(\sigma_z-i\sigma_y)\begin{pmatrix} a \\ b\end{pmatrix}_{i_x,(j+1)_y} + \text{h.c.} \\ &+\sum_{i_x,j_y}\left(a^\dagger,\ b^\dagger\right)_{i_x,j_y}\frac{1}{2}(\sigma_z-i\sigma_x)\begin{pmatrix} a \\ b\end{pmatrix}_{(i+1)_x,j_y} + \text{h.c.}\end{aligned} \tag{10}$$

Now say that one needs m/n sites along x/y direction, define such matrix

$$mr = \begin{pmatrix} 0 & 1 & & & \\ 1 & 0 & 1 & & \\ & 1 & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & 1 & 0\end{pmatrix}_{m\times m},\ mi = \begin{pmatrix} 0 & i & & & \\ -i & 0 & i & & \\ & -i & \ddots & \ddots & \\ & & \ddots & \ddots & i \\ & & & -i & 0\end{pmatrix}_{m\times m}.$$

After careful consideration, one may expects the Hamiltonian to be like

$$\begin{aligned} H =& \mathbb{I}_m \otimes H_y + m_r \otimes (\mathbb{I}_n \otimes \mathrm{Re}T_x) + mi \otimes (\mathbb{I}_n \times \mathrm{Im}T_x) \\ =& \begin{pmatrix} H_y & h_0 & & & \\ h_0^\dagger & H_y & h_0 & & \\ & h_0^\dagger & \ddots & \ddots & \\ & & \ddots & \ddots & h_0 \\ & & & h_0^\dagger & H_y \end{pmatrix}_{m\times n\times \dim(H), m\times n\times \dim(H)} \end{aligned} , \tag{11}$$

where $h_0 = \mathbb{I}_n \otimes T_x$, and T_x is minimal x hopping matrix and equals $\frac{1}{2}(\sigma_z - i\sigma_x)$ in this model, and H_y is $n \times n$ hopping matrix for y direction, which is just like halp-open case along y, but without k_x related terms.

Since it comes to 2-D, now the topological index becomes non-vanishing 1st Chern number

$$\mathrm{Ch}_1 = \int_{BZ^{d=2}} \mathrm{ch}_1(\mathcal{F}),\ \mathrm{ch}_1(\mathcal{F}) = \mathrm{Tr}\Big(\frac{\mathcal{F}}{2\pi}\Big), \tag{12}$$

where the non-Abelian Berry curvature for the only occupied band state $u(k)$ here becomes Abelian and may be defined as

$$\mathcal{F}(k) = \mathrm{d}\mathcal{A}(k),\ \mathcal{A}(k) = i\,\langle u(k)|\,\mathrm{d}\,|u(k)\rangle\,, \tag{13}$$

although such definition seems to provide an extra minus sign compared with mathematical side. Essentially note that one should never directly numerically calculate the Chern number by discrete integration, because when diagonalizing the Hamiltonian at different k, the program usually won't make the wavefunction smoothly connected, i.e., they differ by a local gauge transformation, and thus making the $\mathcal{F}$ in different gauge choice at different k, which is unacceptable. Of course one could analytically integrate it or use special program to generate wavefunction with fixed local gauge, however, it is more convenient to use $U(1)$ link variable [2]

$$U_\mu(k) = \langle u_k | u_{k+k_\mu}\rangle \,/\, |\,\langle u_k | u_{k+k_\mu}\rangle\,|, \tag{14}$$

which equals

$$\gamma_{k,k_\mu} = e^{-i\theta_{k,k_\mu}},\ \theta_{k,k_\mu} = -\arg(\langle u_k | u_{k+k_\mu}\rangle),$$

to generate discrete berry connection and curvature

$$(\mathcal{A})_\mu(k) = i\ln U_\mu(k),\ (\mathcal{F})_{\mu\nu}(k) = i\ln\big(U_\mu(k)U_\nu(k+k_\mu)U_\mu^{-1}(k+k_\nu)U_\nu^{-1}(k)\big). \tag{15}$$

Basically, the function inside 'ln' is like

$$\langle u_k | u_{k+k_\mu}\rangle\,\langle u_{k+k_\mu} | u_{k+k_\mu+k_\nu}\rangle\,\langle u_{k+k_\mu+k_\nu} | u_{k+k_\nu}\rangle\,\langle u_{k+k_\nu} | u_k\rangle\,,$$

except for some renormalization factor, which is ignorable as they contribute to the imaginary part of Chern number. Clearly, this is a loop multiplication which does not care about the gauge and is, thus, gauge invariant. Then, the Chern number could be calculated by

$$c_1 = \frac{1}{2\pi}\sum_k \mathrm{Tr}\left[(\mathcal{F})_{\mu\nu}(k)\right]. \tag{16}$$

For electrical conductivity, though it is directly related to Chern number, the Kubo formula method here is used

$$\sigma_{xy} = -i\frac{e^2}{\hbar}\sum_{E_a<\mu_F<E_b}\int_{BZ}\frac{\mathrm{d}k_x\mathrm{d}k_y}{(2\pi)^2}\frac{\langle b|\frac{\partial H}{\partial k_x}|a\rangle\langle a|\frac{\partial H}{\partial k_y}|b\rangle - \langle b|\frac{\partial H}{\partial k_y}|a\rangle\langle a|\frac{\partial H}{\partial k_x}|b\rangle}{(E_a-E_b)^2} = \frac{e^2}{h}c_1. \tag{17}$$

Notice here on the derivative on k vectors, as one may simply replace $\sin kx \to \cos kx$ and $\cos kx \to -\sin kx$, but forget to delete other terms unrelated. Also, the calculation may be simplified by appropriately using different matrix multiplication, especially multiplication by elements for the division by energy difference. Notice that ε_F as chemical potential is a manually set variable. Also notice that sometimes people would like to consider a thin film geometry, at which time the so-called 'layer Chern number' would be useful. Basically what we need to do is just rewritten. By introducing layer-projection operator

$$|a(l)\rangle = P_l|a\rangle\,,\ \ \sum_l P_l = \mathbb{I},$$

we have

$$\begin{aligned}\sigma_{xy} &= \frac{e^2}{h}\sum_{E_a<\mu_F<E_b}\int_{BZ}\frac{\mathrm{d}^2k}{\pi}\Im\left[\frac{\langle b|\frac{\partial H}{\partial k_x}|a\rangle\langle a|\frac{\partial H}{\partial k_y}|b\rangle}{(E_a-E_b)^2}\right]\\ &= \sum_{l,l'}\frac{e^2}{h}\sum_{E_a<\mu_F<E_b}\int_{BZ}\frac{\mathrm{d}^2k}{\pi}\Im\left[\frac{\langle b(l)|\frac{\partial H}{\partial k_x}|a(l)\rangle\langle a(l')|\frac{\partial H}{\partial k_y}|b(l')\rangle}{(E_a-E_b)^2}\right]\\ &= \frac{e^2}{h}\sum_l c_1(l),\end{aligned}$$

where we have assumed that position-dependent velocity operator is always diagonal, so that we only have two position indices to sum instead of four. The physical meaning for $c_1(l)$ is quite clear as it counts influence from all other layers, i.e., $j_y(l')$ to the lth layer, i.e., $j_x(l)$.

For local density of states (LDOS). For this question, what we want to see is a 2-D color picture, where color represents magnitude of LDOS, varying with one say k_x and proposed energy E (equivalently ω). The definition is just

$$\rho(\omega) = \sum_n \delta(\omega-\omega_n) = -\frac{1}{\pi}\Im\lim_{\varepsilon\to 0^+} G_{00}(\omega), \tag{18}$$

where retarded Green's function is defined as

$$G_{00}(\omega) = \frac{1}{\omega + i\varepsilon - H_{00}}, \tag{19}$$

and such a relation is possible due to Dirac identity as

$$\frac{1}{x + i\varepsilon} = \mathrm{P.V}\frac{1}{x} - i\pi\delta(x), \tag{20}$$

where ε goes to zero and P.V for principle value. Here, H_{00} comes from the open-y boundary Hamiltonian as

$$H = \begin{pmatrix} H_{00} & H_{01} & & & \\ H_{10} & H_{11} & H_{12} & & \\ & H_{21} & \ddots & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & \ddots & \end{pmatrix},$$

as presented above.

Now basically there are two ways to evaluate the question. First is fully employing open-boundary condition, as we may H a finite $\dim(H(k_x, i_y)) = n_y \times \dim(H(k_x, k_y))$ one. And instead of using Green's function, we directly expand Dirac's δ-function as Lorentz function

$$\hat{\rho}(k_x, \omega) = \frac{1}{\pi}\sum_n \lim_{\varepsilon\to 0^+} \frac{\varepsilon\,|n\rangle\langle n|}{(\omega - \omega_n)^2 + \varepsilon^2}, \tag{21}$$

and LDOS at each site (i_y surface if imagining) becomes

$$\rho(k_x, \omega, i_y) = \langle\psi(k_x, i_y)|\hat{\rho}(k_x, \omega)|\psi(k_x, i_y)\rangle = \frac{1}{\pi}\sum_n \lim_{\varepsilon\to 0^+} \frac{|\psi_n(k_x, i_y)|^2}{(\omega - \omega_n)^2 + \varepsilon^2}, \tag{22}$$

and all we need to do is to solve eigenstates and eigenvalues of $H(k_x, i_y)$ and bring them above, then plot LDOS of the layer we want to see. Notice that energy set composed by ω could be vectorized so calculation is greatly simplified.

The method above is effective, but somehow costs much and may be inaccurate as y is totally open, and we have second method to directly calculate the surface Green's function based on semi-infinite y geometry using $\dim(H(k_x, k_y))$ matrix to avoid this. There are several methods to solve this semi-infinite problem, and we use the one known as transfer matrix method[3]. Given semi-infinite H above, its Green's function is defined as

$$(\omega - H)G(\omega) = 1, \tag{23}$$

where 1 is semi-infinite identity matrix which one may make clear by oneself. Similarly, we have

$$G = \begin{pmatrix} G_{00} & G_{01} & G_{20} & & \\ G_{10} & G_{11} & G_{12} & & \\ G_{20} & G_{21} & \ddots & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & \ddots & \end{pmatrix}.$$

To get G_{00}, we only set of equations based on first column. By observation, we have

$$\begin{cases} (\omega - H_{00})G_{00} - H_{01}G_{10} = I \\ (\omega - H_{nn})G_{n0} = H_{n,n-1}G_{n-1,0} + H_{n,n+1}G_{n+1,0} \end{cases}, \tag{24}$$

where one may notice that

$$H_{ii} = H_{00},\ H_{n,n+1} = H_{01}, H_{n,n-1} = H_{10} = H_{01}^{\dagger}.$$

By defining

$$\begin{cases} t_0 = (\omega - H_{00})^{-1}H_{01}^{\dagger} \\ \tilde{t}_0 = (\omega - H_{00})^{-1}H_{01} \end{cases}, \tag{25}$$

one may find iteratively that

$$\begin{aligned} G_{n0} &= t_0 G_{n-1,0} + \tilde{t}_0 G_{n+1,0} \\ &= t_0(t_0 G_{n-2,0} + \tilde{t}_0 G_{n,0}) + \tilde{t}_0(t_0 G_{n,0} + \tilde{t}_0 G_{n+2,0}), \end{aligned}$$

which leads to

$$G_{n0} = t_1 G_{n-2,0} + \tilde{t}_1 G_{n+2,0}, \tag{26}$$

by

$$\begin{cases} t_1 = (I - t_0\tilde{t}_0 - \tilde{t}_0 t_0)^{-1}t_0^2 \\ \tilde{t}_1 = (I - t_0\tilde{t}_0 - \tilde{t}_0 t_0)^{-1}\tilde{t}_0^2 \end{cases}. \tag{27}$$

This total iterative form could again be generalized as

$$G_{n0} = t_i G_{n-2^i,0} + \tilde{t}_i G_{n+2^i,0}, \tag{28}$$

where 2^i is actual exponential form, with coefficients

$$\begin{cases} t_i = (I - t_{i-1}\tilde{t}_{i-1} - \tilde{t}_{i-1}t_{i-1})^{-1}t_{i-1}^2 \\ \tilde{t}_i = (I - t_{i-1}\tilde{t}_{i-1} - \tilde{t}_{i-1}t_{i-1})^{-1}\tilde{t}_{i-1}^2 \end{cases}. \tag{29}$$

Now say $n = 2^i$, and we have

$$G_{2^i,0} = t_i G_{00} + \tilde{t}_i G_{2^{i+1},0},$$

which again provides us with another iterative form starting from $i = 0$:

$$\begin{aligned} G_{10} &= t_0 G_{00} + \tilde{t}_0 G_{20},\ G_{20} = t_1 G_{00} + \tilde{t}_1 G_{2^2,0} \\ &= (t_0 + \tilde{t}_0 t_1) G_{00} + \tilde{t}_0 \tilde{t}_1 G_{2^2,0} \\ &= \cdots \\ &= T_i G_{00} + \prod_{j=0}^{i} \tilde{t}_j G_{2^{i+1},0}, \end{aligned}$$

where ith-order transfer matrix is defined as

$$T_i = \sum_{j=0}^{i} \tilde{t}_0 \tilde{t}_1 \cdots \tilde{t}_{j-1} t_j = t_0 + \tilde{t}_0 t_1 + \tilde{t}_0 \tilde{t}_1 t_2 + \cdots. \tag{30}$$

By assuming that

$$\lim_{i \to +\infty} \prod_{j=0}^{i} \tilde{t}_j = 0,$$

we define

$$T = \lim_{i \to +\infty} T_i \tag{31}$$

as transfer matrix and write

$$G_{10} = T G_{00}, \tag{32}$$

bring which back to the first equation in set we see

$$G_{00} = \frac{1}{\omega - H_{00} - H_{01} T}. \tag{33}$$

This method is faster but could only calculate first (or last) layer LDOS.

Speaking of last layer, one may notice that at this time,

$$H = \begin{pmatrix} \ddots & \ddots & & & \\ \ddots & \ddots & \ddots & & \\ \ddots & & H_{N-2,N-1} & & \\ & H_{N-1,N-2} & H_{N-1,N-1} & H_{N-1,N} \\ & & H_{N,N-1} & H_{NN} \end{pmatrix},\ G = \begin{pmatrix} \ddots & \ddots & & & \\ \ddots & \ddots & \ddots & & \\ \ddots & & G_{N-2,N-1} & G_{N-2,N} & \\ & G_{N-1,N-2} & G_{N-1,N-1} & G_{N-1,N} & \\ & G_{N,N-2} & G_{N,N-1} & G_{NN} & \end{pmatrix},$$

where we manually identify $N = +\infty$. Then similarly, one could get iterate equations using last column of G

$$\begin{cases} (\omega - H_{00})G_{NN} - H_{01}^{\dagger} G_{N-1,N} = 1 \\ (\omega - H_{00})G_{nN} = H_{01}^{\dagger} G_{n-1,N} + H_{01} G_{n+1,N} \end{cases}, \tag{34}$$

and

$$G_{nN} = t_i G_{n-2^i,N} + \tilde{t}_i G_{n+2^i,N}, \tag{35}$$

with the same definition above. However, now we utilize $n = N - 2^i$ as

$$G_{N-2^i,N} = t_i G_{N-2^{i+1},N} + \tilde{t}_i G_{NN}, \tag{36}$$

then also start from $i = 0$, we similarly have

$$G_{N-1,N} = \cdots = \prod_{j=0}^{i} t_0 t_1 \cdots t_i G_{N-2^{i+1},N} + \tilde{T}_i G_{NN}, \tag{37}$$

where the new

$$\tilde{T}_i = \sum_{j=0}^{i} t_0 t_1 \cdots t_{i-1} \tilde{t}_i = \tilde{t}_0 + t_0 \tilde{t}_1 + t_0 t_1 \tilde{t}_2 + \cdots. \tag{38}$$

Also similarly, we recognize first item to be zero and let

$$\tilde{T} = \lim_{i \to +\infty} \tilde{T}_i, \tag{39}$$

then we have

$$G_{NN} = \frac{1}{\omega - H_{00} - H_{01}^{\dagger} \tilde{T}}. \tag{40}$$

However, notice that Green function here contains no real-space information of the surface, and if one wants to know that (say 3-D TI thin film, z is finite, x is periodic and y is semi-infinite, and one wants to get LDOS varying with z at $y = 0$), one may consider the effective Hamiltonian on the (front) surface

$$H_s(k_x, \omega) = H_{00}(k_x) + H_{01}(k_x) T(k_x, \omega), \tag{41}$$

then based on this, one may calculate z-dependent LDOS

$$\rho(k_x, \omega, z) = \frac{1}{\pi} \sum_n \lim_{\epsilon \to 0^+} \frac{|\psi_n(k_x, \omega, z)|^2}{(\omega - \omega(k_x, \omega))^2 + \epsilon^2}, \tag{42}$$

where except for k_x, eigen-information depends on ω, too. However, this is just the understanding, and in real practice, one notice that the effective Hamiltonian justs accounts for G_{00}, and the position information is already included in as long as it is clear in H_{00}.

Except for LDOS, another position dependent quantity local current density or local density of current (LDOC) is also important. Its calculation is quite simple upon definition. Using j_x as example, since

$$j_x = ev_x = \frac{e}{\hbar}\partial_{k_x} H,$$

one recognizes LDOC as its position dependent expectation value as

$$j_x(r) = \langle\phi(r)|j_x|\phi(r)\rangle = \frac{e}{h}\int \mathrm{d}k_x \sum_n \langle\phi_n(r)|\partial_{k_x} H|\phi_n(r)\rangle\,, \tag{43}$$

where eigenstates could be solved using exact diagonalization upon open boundary.

2. Honeycomb Lattice

By this, we mean the Haldane's model with extra mass term on different kinds of sites, and finally, on different cones.

$$\begin{aligned}
\hat{H} &= \hat{H}_{NN} + \hat{H}_{NNN} + m\sum_i (a^\dagger_{r_i} a_{r_i} - b^\dagger_{r_i} b_{r_i}),\\
\hat{H}_{NN} &= -t\sum_i \sum_{j=1}^{3} (a^\dagger_{r_i} b_{r_i+e_j} + \text{h.c.})\\
&= \sum_k \begin{pmatrix} a^\dagger_k & b^\dagger_k \end{pmatrix} \begin{pmatrix} & -t\sum_j e^{-ik\cdot e_j} \\ -t\sum_j e^{ik\cdot e_j} & \end{pmatrix} \begin{pmatrix} a_k \\ b_k \end{pmatrix},\\
\hat{H}_{NNN} &= -t'\sum_i \sum_{j=1}^{3} (e^{i\phi} a^\dagger_{r_i} a_{r_i+v_j} + e^{-i\phi} b^\dagger_{r_i} b_{r_i+v_j} + \text{h.c.})\\
&= \sum_k \begin{pmatrix} a^\dagger_k & b^\dagger_k \end{pmatrix} \begin{pmatrix} -2t'\sum_j \cos\big(k\cdot v_j - \phi\big) & \\ & -2t'\sum_j \cos\big(k\cdot v_j + \phi\big) \end{pmatrix} \begin{pmatrix} a_k \\ b_k \end{pmatrix},
\end{aligned} \tag{44}$$

where e_j & v_j denote to the nearest-neighbor and next-nearest-neighbor lattice vectors, respectively.

The energy band ploting in k-space is common, however, when one tries to explore the edge state, one finds two different edges, say, zigzag and armchair. And one needs to be very careful after performing the half-Fourier transformation, which is time to write down real-space-open-boundary Hamiltonian. That is because, the periodic axis is still under Fourier transformation and is assigned with k-space quantum number, and one should be reminded that all information about this axis in real space has already gone, under such consideration, one may meet the case that along open direction, hopping is no longer simply to the nearest site, and it is the time one needs to expand the periodicity along the direction. In fact, one can just align atoms along the open direction, since the Fourier transformation will take care of other periodic directions. More directly, one just do the anti-Fourier transformation and see what happens.

For this honeycomb model, one may find that zigzag edge needs no expansion while armchair edge needs. What is more, zigzag open boundary shows zero modes while armchair does not. This is simply due to the formation of a non-trivial 1-D model along the opened direction, which gives rise to zero mode on boundary. Here, by choosing $e_1 = (0, a)$, $e_2 = (-\sqrt{3}a/2, a/2)$, $e_3 = (\sqrt{3}a/2, a/2)$, the NN graphene Hamiltonian reads

$$H_{NN} = \left[2\cos\left(\frac{\sqrt{3}k_x}{2}\right) + \cos\left(\frac{k_y}{2}\right)\right]\sigma_x + \sin\left(\frac{k_y}{2}\right)\sigma_y, \tag{45}$$

with $a = 1$ and a factor absorbed into fermion operators. The armchair condition opens along x, then since $H_{NN}(k_x, k_y \equiv Q)$ is always trivial for all Q, there is no zero mode. The zigzag condition opens along y, and non-trivial $H_{NN}(k_x \equiv Q, k_y)$ requires $|\cos(\sqrt{3}Q/2)| < 1/2$, then there will be zero modes when $\pi/3 < |\sqrt{3}k_x/2| < 2\pi/3$.

C. 3-D

Here, we consider the simplest 3-D TI model, modified by Zeeman field on z-direction and thus change from $\mathbb{Z}_2$ to $\mathbb{Z}$, and in fact, from class CII to class AIII.

$$H(k) = A\sum_{i=1}^{3}\sin k_i\alpha_i + (m - 4B\sum_{i=1}^{3}\sin^2\frac{k_i}{2})\beta + Z\tau_0\otimes\sigma_z, \tag{46}$$

where $\alpha_i = \tau_x\otimes\sigma_i$, and $\beta = \tau_z\otimes\sigma_0$, with τ and σ represent for orbital and spin degrees of freedom, respectively.

For such a 3-D system, it is impossible to draw its energy band visibly in our space, thus we can only cut for one k-space cross section and plot for that. However, now one is allowed to draw edge state with open boundary on one direction using 3-D plot and sees the edge Dirac cone. Here, abandoning the Zeeman term and consider any direction to be the boundary, one would see that since another two axises are assigned with good quantum k numbers, their contribution to edge state is some kind of trivial as virtically waves. And one can always reduce the Hamiltonian as a 1-D one

$$H = A\sin k_i\alpha_i + (m - 4B\sin^2\frac{k_i}{2})\beta,$$

whose spinor part is clearly eigenstates of the multiplication $\alpha_i\beta$, which also contains the chiral operator of this 1-D Hamiltonian, as one can verify easily by multiplying α_i on the left of eigenfunction.

Basically, the main problem may come from the calculation of Chern number or the Hall conductivity, when considering z open boundary and treat the system as a quasi-2D one. Since it is

now a thin film rather than one atomic layer, there are many bands and some would have so-called accidental degeneracy. Well, for the system, it is accidental, but for our consideration, this means that we have to use non-Abelian berry connection and curvature to go forward. Luckily, one may generalize (14) as

$$U_\mu(k) = \frac{1}{N_\mu(k)} \det\big(\Psi^\dagger(k)\Psi(k+k_\mu)\big), \tag{47}$$

where $N_\mu(k) = |\det\big(\Psi^\dagger(k)\Psi(k+k_\mu)\big)|$, and $\Psi(k) = (|u_{1,k}\rangle, \cdots, |u_{n,k}\rangle)$. And for conductivity calculation using Kubo formula, we may make the nominator as $(\epsilon_a - \epsilon_b + i\delta_+)^2$ to avoid the divergence problem.

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